Secretory protein pathway

Follow a soluble secreted protein without confusing information flow with cargo flow.

Information route

DNA in nucleus → mRNA through a nuclear pore → Ribosome translates mRNA

Cargo route for a soluble secreted protein

Rough ER → Transport vesicle → Golgi apparatus → Secretory vesicle → Plasma membrane → Extracellular fluid

Location	What happens here
Ribosome at rough ER	An ER-targeting signal directs the nascent chain into the pathway; the polypeptide is assembled by a ribosome.
ER -> Golgi	Cargo is transported in membrane carriers; processing and sorting continue in Golgi compartments.
Vesicle fusion	The lumen opens to the exterior. A soluble lumenal protein is released during exocytosis.

Common pitfall: the Golgi does not translate mRNA. Not every protein goes through this route; many cytosolic proteins remain outside the ER.

Secretory protein pathway

Self-check questions and explanations. Refer to the visual guide on the front.

What route does a soluble secreted protein follow?

ER -> transport vesicle -> Golgi -> secretory vesicle -> plasma membrane -> extracellular fluid. The precise Golgi transport mechanism is beyond this sheet.

What leaves the nucleus as information for protein synthesis?

mRNA leaves through nuclear pores and is translated by ribosomes. DNA stores sequence information; it is not physically converted into a protein or exported as one.

Does the Golgi assemble the amino-acid sequence?

Ribosomes at rough ER synthesize the polypeptide in this case. The Golgi modifies, sorts and routes cargo; it does not translate an mRNA into its amino-acid sequence.

What happens when ER-to-Golgi transport is blocked?

Cargo can accumulate in the ER or associated blocked transport compartments. Less newly made cargo reaches the Golgi and secretory vesicles; the exact pattern depends on the location of the block.

Where does soluble vesicle cargo go during exocytosis?

The enzyme is released outside. Membrane fusion connects the vesicle lumen with the extracellular space while maintaining membrane orientation; the bilayer is not turned inside out.

Must a cytosolic enzyme pass through ER and Golgi?

No. A protein functioning in cytosol can be translated on free ribosomes and remain there. An ER-targeting signal sequence, recognized by targeting machinery, directs appropriate nascent proteins to ER.